

Analyzing Gamification in E-learning tools : HCI-Driven Insights from University Students in Bangladesh

Abstract

Gamification has emerged as a widely adopted strategy to enhance learner motivation and engagement in online education, yet its impact in South Asian contexts remains underexplored. This study investigates how gamification features in e-learning platforms influence learner engagement among university students in Bangladesh. A survey of 218 students across multiple universities was conducted to capture perceptions of gamified features and expectations of an ideal learning system. The study identified seven key gamification elements: interactive tasks, lesson completion rewards, leaderboards, achievement animations, task completion discounts, free preview lessons, and detailed course outlines—and examined their prevalence across 24 platforms commonly used in Bangladesh. A within-subjects design ensured consistency, and statistical analysis, including heatmap visualizations, revealed that top-ranked platforms consistently integrated core features such as interactive tasks, rewards, previews, and course outlines. In contrast, low and mid-tier platforms lacked such features, relying instead on minimal transparency mechanisms. Findings validate the hypothesis that lack of gamification significantly reduces engagement, while also highlighting mismatches between student expectations and platform design. This study contributes to the literature by providing context-specific, HCI-driven insights into gamification in Bangladesh and offers evidence-based guidance for the design of culturally relevant and student-centered e-learning systems.

Keywords

Gamification, E-learning, Human-Computer-Interaction (HCI).

1 Introduction

The integration of digital technologies into higher education has transformed the way students access and interact with learning content. Learning management systems (LMS) such as Moodle, Blackboard, and Canvas are now widely used to deliver instructional materials, monitor progress, and facilitate communication between teachers and students [1]. However, despite the accessibility and scalability of such platforms, several challenges persist. Students often report low levels of motivation, limited engagement, and difficulties in maintaining long-term participation, which in turn contributes to higher dropout rates [2].

Gamification has been proposed as a promising approach to address these issues. It refers to the use of game-like features—such as points, leaderboards, badges, and rewards—in non-game contexts with the goal of enhancing motivation and participation. Within e-learning environments, these elements seek to transform routine learning activities into interactive experiences that can foster a sense of achievement, competition, and progression [3]. Prior studies indicate that gamification can improve student engagement, increase time-on-task, and positively influence academic performance [4].

From a Human-Computer Interaction (HCI) perspective, the effectiveness of gamification depends not only on the availability of features but also on how learners perceive, interact with, and adapt to them in their cultural and educational contexts [5]. For example, while some learners respond positively to competitive mechanisms like leaderboards, others may prefer collaborative elements or personalized feedback. This indicates the need for context-specific investigations that consider both user experience and cultural adaptation in the design of gamified platforms.

In Bangladesh, the adoption of e-learning has accelerated due to increasing digitisation in higher education, yet challenges related to motivation and engagement remain significant [6]. Despite the presence of global LMS platforms, most are not tailored to the local linguistic and cultural environment, limiting their effectiveness for Bangladeshi students.

To address these challenges, our study explores the integration of gamification in Bangladeshi online-learning platforms, focusing on the perceptions and experiences of students in Bangla-language contexts. By combining survey-based data with HCI-driven analysis, it aims to identify which gamified features are most effective in sustaining motivation, improving engagement, and supporting academic performance. The findings will not only contribute to the growing body of literature on gamification in education but also provide design recommendations for culturally relevant and student-centred online-learning systems.

The remaining part of this paper is structured as follows: Section 2 delineates the problem statement, whereas Section 3 elucidates the pertinent literature associated with our research. Section 4 delineates the research gaps. The methodology is detailed in section 5, while the experimental results are reported in part 6. The findings of this investigation are given and analyzed in Section 7. Sections 8, 9, and 10 conclude the paper by addressing the limitations and future scope of our research.

2 Problem Statement

Active participation in e-learning platforms is crucial for maintaining student motivation and enhancing learning results. Nevertheless, numerous platforms in Bangladesh inadequately utilise or inconsistently implement gamification features, resulting in diminished involvement, insufficient incentive, and elevated dropout rates. Our research examines the most effective gamification aspects for Bangladeshi learners and explores how context-specific, HCI-driven design might improve engagement, cultural relevance, and academic achievement.

2.1 Hypothesis Statement

We propose that the lack or irregular application of gamification features in Bangladeshi e-learning platforms markedly diminishes learner motivation and engagement, while the methodical incorporation of contextually pertinent gamification elements can augment participation and enhance learning outcomes.

2.2 Research Questions

To address this problem statement, we explore the following research question:

- (1) What gamification features (such as points/badges, leaderboards, interactive challenges, or previews) are most effective in improving motivation and learning outcomes for Bangladeshi students?
- (2) To what extent do demographic factors such as year of study, field of study, and age group influence students' preferences for gamification features—ranging from competitive elements (leaderboards, rewards, discounts, achievement animations) to collaborative and trust-building features (interactive tasks, detailed course outlines, and free previews)—in Bangladeshi e-learning platforms?
- (3) How do gamification features vary across leading versus lower-tier platforms, and what impact do these differences have on overall platform engagement?

3 Related Works

A promising strategy to improve learning is gamification, which involves incorporating game-like features into educational systems, such as leaderboards, badges, points, and feedback. Its efficacy in raising academic achievement, motivation, and engagement has been studied over the last ten years. Four recent studies that shed light on the advantages and difficulties of gamification in education are reviewed in this section.

In a broader systematic review of gamification in e-learning (2014–2023) [7] synthesized findings from 37 studies to examine both benefits and challenges. The review concludes that gamification generally improves motivation, engagement, and learning effectiveness. Students particularly benefited from immediate feedback, progress tracking, and recognition through rewards. Teachers also reported advantages in terms of easier progress monitoring and increased classroom interaction. However, the review highlights several challenges, including poor alignment of game design with pedagogy, lack of personalization, and teacher resistance to adopting gamified methods.

Ahmed et al. [8] adopted a more empirical approach, testing the effects of gamification in a Moodle-based learning environment with 192 students. Their quasi-experimental design demonstrated a statistically significant improvement in academic performance after gamified components such as points, badges, and leaderboards were introduced. The results confirm that gamification can have a measurable positive impact on learning outcomes. Nonetheless, the study focused primarily on performance metrics and did not account for other factors such as student motivation or long-term retention.

Bulbulia et al. [9] explores how competitive gamification can improve motivation and performance in software testing education in their work. They address the reluctance of students toward testing by gamifying test case writing through a tool called Codetierlist. Codetierlist ranks student code on a leaderboard by comparing their tests and solutions with peers and instructor solutions. The evaluation in a first year Computer Science course showed students produced more correct code, better test cases and even identified bugs in instructor code. However, while motivation increased, the

self-efficacy did not improve and some low ranked students felt discouraged due to this system. They presented a publicly available gamified testing platform and insights into balancing competition with student well-being.

Jun and Lucas [10] conducted a systematic review of studies published between 2021 and 2024, focusing on how different game elements affect student outcomes. Their review highlights that gamification significantly improves motivation, active participation, and academic performance. Badges and leaderboards foster competition and recognition, while point systems provide instant feedback and clear goals, making learning more interactive. However, they caution that game elements must be carefully aligned with learning objectives to avoid distraction.

Lampropoulos et al. [11] examined the impact of gamification on higher education learning outcomes compared with online and traditional methods. They also conducted a three year longitudinal study with 1001 students in a computer science course. Their methodology included quantitative analysis of grades, success, excellence, withdrawal, engagement and motivation which is supported by a 20 item questionnaire. The findings from the study showed gamified learning outperformed both online and traditional formats in success rate, excellence rate, average grade and retention. Moreover, students reported higher motivation, engagement and satisfaction, aligning with previous works on the benefits of gamification. Overall, the study provides strong evidence that gamification enhances academic performance and student experience more effectively than other approaches.

Another research by Meng et al. [12] examines how specific gamification components (Points and Badges) influence the engagement of online learners. The study was motivated by gaps in prior work that mixed multiple components without isolating their individual effects. Using a mixed-method approach, data were collected from 440 learners across 19 countries on the Course of Networking. Their important findings revealed that points significantly boosted the skills, emotional, participation and performance engagement, while Badges only affected participation. The perceptions of students towards gamification were mixed by showing advantages but also some limitations in design and measurement. They highlighted the need to tailor gamification components for students carefully and provides directions for future research from instructor and designer perspectives.

Oliveira et al. [13] reviewed how gamification can foster e-learning and sustainability by organizing the existing fragmented literature into a coherent framework. They applied a systematic literature review using the PRISMA protocol by analyzing 130 Web of Science articles. Their analysis mapped key behavioral attitudes: motivation, engagement and satisfaction induced by gamification. They also identified seven main game elements (user-centricity, challenges, narrative, loops, competition, feedback, rewards) that support active learning. Their study proposed the 7GOALS framework by linking these elements to the PDCA cycle for continuous improvement in e-learning. The authors highlighted the potential of gamification to stimulate sustainable behaviors and enhance learning outcomes across disciplines.

Similarly, Sevilla-Campoverde et al. [14] investigated gamification in graduate-level e-learning using a mixed-methods approach.

They analyzed gamification tools across different platforms and surveyed experts to evaluate their effectiveness. The findings suggest that gamification enhances student engagement, participation, and satisfaction, though results vary across educational levels. Interestingly, combining competitive and collaborative elements showed even greater potential. The study also emphasizes the importance of designing gamification strategies that support intrinsic motivation rather than relying solely on rewards.

This paper by Smirani et al. [15] investigates how gamification elements (Kahoot!, Classcraft, Badgeville) affect engagement, motivation, and knowledge retention in online learning. They address the inconsistent findings in prior studies and clarify the role of specific gamification strategies. The authors used a survey of 169 academics and applied Structural Equation Modelling (SEM) guided by Self-Determination Theory. Findings showed a 25% rise in engagement, 30% performance improvement, and a 20% increase in enjoyment, though some doubts about learning effectiveness persisted. They showed that knowledge retention correlated strongly is 0.65 with engagement and perceived effectiveness. The outcome of the study is evidence based guidance for integrating balanced gamification to enhance e-learning environments.

Taken together, these studies suggest that gamification can be a powerful tool for enhancing learning experiences when applied thoughtfully. While evidence supports its ability to increase engagement and performance, success largely depends on careful integration, alignment with educational goals, and consideration of both extrinsic and intrinsic motivational factors. Future research should explore adaptive designs, long-term impacts, and perspectives of both students and educators to maximize the potential of gamification in education.

4 Research Gaps

Previous research has yielded valuable insights on gamification globally, but several key gaps remain in the Bangladeshi context:

- **Local/Cultural Gap:** Most studies focus on Western or generalized settings, with little examination of how cultural factors in Bangladesh affect gamification efficacy. Bangladeshi learners use predominantly Bengali-language materials and may value different game features than international cohorts. Our work directly addresses this gap by centering on Bangladeshi higher education and student feedback.
- **Platform Scope Gap:** Prior work often analyzes single or a few platforms. In contrast, we review 24 online learning platforms (including 20+ regional systems and major global services) to provide a comprehensive view. This broad scope reveals how platform design varies and which core features are missing in less successful systems.
- **Practical Design Features Gap:** While many papers discuss the benefits of gamification in general, few distill specific design elements that practitioners should implement. We identify seven practical gamification features that Bangladeshi students expect (e.g. small interactive tasks, lesson-completion rewards, free previews etc.). These concrete features can guide designers in creating engaging interfaces.

- **Proof of Implementation Gap:** By analyzing platform UI, we show which systems include or omit each feature. This “proof of implementation” highlights mismatches between student expectations and actual designs, informing targeted improvements.

5 Our Approach

To evaluate the correlation between gamified features and learner engagement in online platforms, it is essential to identify the features that are most suitable and evolutionary. Once these features are determined, it becomes feasible to examine their presence across platforms of varying levels of popularity. However, the process of feature shortlisting is cumbersome, as it requires considering the perspectives of diverse learners. Therefore, our approach consists of the following steps:

- (1) Explore a wide range of online platforms to compile a comprehensive list of potential features that incorporate gamification strategies and hold practical value for learners.
- (2) For the purpose of data collection, design a form that effectively captures the expectations of a broad audience from their own perspective. The form should clearly reflect participants’ views on the inclusion of our selected gamified features in online learning tools, while also recording the popularity of different platforms among participants.
- (3) Distribute the form and collect responses. Based on the data, rank the gamified features and establish a shortlist of the most preferred ones. Additionally, rank and categorize the platforms according to user responses, and store the processed data for further analysis.
- (4) Apply statistical procedures to the collected data across institutions of varying ranks in order to test the hypothesis and determine whether it can be supported or refuted.

6 Experiments

We formulated curated features based on responses from a survey conducted among a large participant group regarding their perspectives on gamification and intuitive interface. Besides, we assessed the overall engagement and popularity of the platforms using the participants’ responses. This approach on the same participants ensured consistency in the experimental findings. Subsequently, statistical analysis was performed to deduce the relationship between the identified features and engagement. The following sections describe our experiments in detail.

6.1 Participants

In order to relate the expectation of learners to the experiences provided by online learning platforms in Bangladesh, we required the perspective of the actual users of these systems. People doing their undergraduate or graduate degrees are the most appropriate participants in our particular case. Therefore, we opted for the current university-goers as our subject. We conducted a survey among 218 people to capture their expectation from a perfect online learning system. The survey continued for two weeks. The specifics of the participants are shown in Table 1.

Table 1: Demographic distribution of survey participants

Variable	Category	Percentage (%)
Year of Study	1st Year	23
	2nd Year	33
	3rd Year	24
	4th Year	11
	MSc/Graduate	9
Major Field	CSE	78
	EEE	14
	Others	8
Age Group	20–24 yrs	89
	25–29 yrs	7
	<20 yrs	4

6.2 Research Variables

We conducted the survey through a Google Form, which was randomly circulated among more than ten universities in Bangladesh. This ‘Form’ method was selected as online questionnaires provide an efficient means of reaching a large and diverse user group while enabling rigorous analysis of the collected responses. To design the form, it was first necessary to identify the original UI features available in existing platforms. For this purpose, we reviewed more than twenty learning platforms, along with widely used international systems such as Udemy, Coursera, and Brilliant.org. After compiling all the possible features observed in the platforms, we transformed them into a series of survey questions. The finalized questionnaire was organized into a six-page form consisting of 24 questions, which was then distributed to participants. Based on the percentage of students expecting particular features, we short-listed seven major features, as shown in Table 2. These features were treated as the independent variables in our analysis, while the engagement of the systems, recorded from the same survey, was considered as the dependent variable.

Table 2: Considered gamified features and percentage of students expecting them

Variable Type	Name of the Gamification Feature	Students Expected (%)
Independent, Binary	Small Interactive Tasks	90.4
	Lesson Completion Rewards	84.9
	Progress Leaderboard	83.9
	Achievement Animations	79.8
	Task Completion Discounts	82.6
	Free Preview Lessons	93.6
	Detailed Course Outline	94.0
Dependent	Popularity / Engagement	–

6.3 Experimental Design

We employed a within-subjects procedure for the survey, in which all participants responded to an identical set of questionnaires presented in the same order. This design ensured that each participant experienced approximately the same conditions while completing

the form. Consequently, the resulting responses and subsequent analysis were rendered more consistent and reliable.

Table 3: Institutions by Rank Group with Popularity

Group	Platform	Popularity
Top-Tier	10 Minute School	1
	Programming Hero	2
	Coursera	3
	Udemy	4
	Ostad	5
High-Tier	Bohubrihi	6
	UpSkill Bangladesh	7
	CodersTrust Bangladesh	8
	Brilliant.org	9
	Skill Hub Bangladesh	10
Middle-Tier	Shikhbe Shobai	11
	AmarSkill	12
	MSB Academy	13
	Opedemy	14
	CPS Academy	15
Low-Tier	Maslow Bangladesh	16
	Coder Vai	17
	DreamLand IT	17
	Interactive Cares	17
	Pondit	17

6.4 Statistical Measures

The platforms were categorized into four mutually exclusive classes: Top-tier, High-tier, Mid-tier, and Low-tier, based on their popularity among the students. The platform-specific statistics is demonstrated in Table 3. This categorization reflects the actual enthusiasm of learners for these platforms, effectively representing their engagement. Also, the foreign-based platforms in the Middle-tier and Low-tier have been eliminated to focus only on the Bangladeshi viewpoint, keeping total 17 regional and 3 international platforms for our analysis. Subsequently, the shortlisted gamification features were evaluated against each of the educational platforms, and the availability of each feature was recorded. We then performed a Heatmap Analysis to examine the relationship between gamification features and engagement for each group. The heatmap provides a comprehensive overview in a compact visual form, facilitating the identification of core features that contribute to learner engagement. Our objective was to compare the engagement levels of Top-tier platforms with those of Low-tier platforms, as well as to analyze the presence of the identified features across these categories.

7 Results

This study evaluated 17 e-learning platforms commonly used in Bangladesh, collecting feedback from 218 university students. This section delineates the outcomes of our survey and platform analysis, addressing the research questions and substantiating the hypothesis concerning the function of gamification in Bangladeshi e-learning platforms.

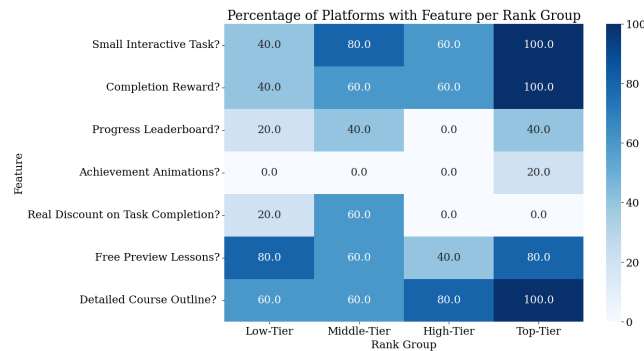
Table 4: Importance of Different Features in E-learning Platforms Based on Student Expectations and Presence in Top-ranked Platforms.

Feature Name	% BD Students Expecting This Feature	Observation/Comment	Importance Based on Top Platforms
Small Interactive Tasks	90.4	Present in top-engagement platforms; absent in low/mid-ranked	Important
Lesson Completion Rewards	84.9	Present in top-engagement platforms; absent in low/mid-ranked	Important
Progress Leaderboard	83.9	Absent in low/mid-ranked platforms; less effect on engagement	Less Important
Achievement Animations	79.8	Absent in low/mid-ranked platforms; less effect on engagement	Less Important
Task Completion Discounts	82.6	Absent in all platforms	Not Important
Free Preview Lessons	93.6	Present in top-engagement platforms; absent in low/mid-ranked	Important
Detailed Course Outline	94.0	Present in top-engagement platforms; absent in low/mid-ranked	Important

7.1 Most Effective Gamification Features (RQ1)

Table 4 demonstrates that students prioritise small interactive assignments (90.4%), detailed course outlines (94.0%), and complimentary preview lessons (93.6%). Rewards for lesson completion (84.9%) were likewise deemed highly significant. These attributes were constantly evident in high-tier platforms (Figure 2, Figure 1) but were predominantly lacking in low- and mid-tier platforms. Conversely, elements like progress leaderboards (83.9%) and achievement animations (79.8%), while appreciated by students, demonstrated a diminished impact on engagement within current platforms, indicating they serve a complementary rather than a principal function. Task completion discounts (82.6%) were nonexistent across all platforms, signifying minimal significance for sustained participation.

Answer to RQ1: Our findings show that small interactive tasks, free preview lessons, detailed course outlines and lesson completion rewards are most efficacious in maintaining motivation and enhancing learning outcomes for Bangladeshi students.

**Figure 1: Feature distribution across e-learning platforms by rank group**

7.2 Demographic Influences on Gamification Preferences (RQ2)

To address RQ2, we conducted Chi-square tests to examine associations between demographic factors (year of study, field of study, and age group) and gamification feature preferences. Responses were simplified into Yes/No categories, and demographic groups were collapsed into broader categories (Undergraduate vs. Graduate, CSE vs. Other majors, and 24 vs. >24 years) to ensure statistical validity.

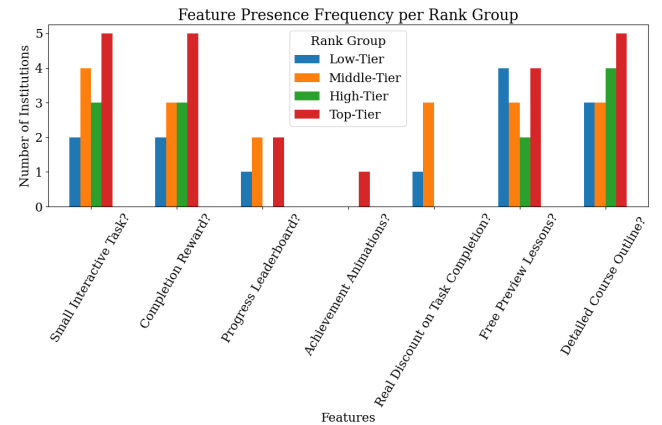
The table 5 revealed no statistically significant subgroup differences (all $p > 0.05$). The lowest observed p-value was 0.293 for Year

of Study $\times$ Course Outline preference ($X^2 = 15.23$), which suggests minor differences across years but not strong enough to confirm statistical significance. Other associations—such as Major $\times$ Leaderboard preference ($X^2 = 12.73$, $p = 0.623$) and Age $\times$ Achievement Animations ($X^2 = 1.88$, $p = 0.597$)—likewise indicated broadly consistent preferences across groups.

Table 5: Chi-square Summary Table with Interpretations

Demographic $\times$ Feature	X^2	p-value	Interpretation
Year $\times$ Course Outline	15.23	0.293	Minor differences
Major $\times$ Leaderboard	12.73	0.623	CSE students slightly higher
Age $\times$ Animations	1.88	0.597	Similar across ages
Year $\times$ Rewards	6.71	0.916	Consistent across years
Major $\times$ Free Previews	12.31	0.655	Similar across majors

Answer to RQ2: Our studies suggest that preferences for both competitive (leaderboards, rewards, discounts, animations) and collaborative/trust-building features (interactive tasks, course outlines, free previews) are largely uniform across demographic groups. The absence of subgroup effects strengthens the interpretation that cultural attitudes toward gamification may play a greater role than individual demographic variables in shaping learner engagement.

**Figure 2: Rank comparison by selected gamification features**

7.3 Feature Importance and Engagement (RQ3)

Figures 1 and 2 demonstrate distinct differences in feature adoption among rank groupings. Top-tier platforms consistently featured

interactive tasks, rewards for lesson completion, complimentary preview lessons, and detailed course outlines, achieving adoption rates of 80–100%. Conversely, low- and mid-tier platforms executed key features inconsistently, frequently achieving adoption rates below 60%. Table 4 corroborates this, indicating that these elements are found in top-tier platforms and are significantly esteemed by students, with over 90% expressing a preference for previews and outlines.

Lower-ranked platforms exhibited a heightened dependence on transactional or superficial incentives. For example, task completion discounts were present in certain low- and middle-tier platforms but completely lacking in high- and top-tier groups (Figure 1). Likewise, achievement animations and progress leaderboards—despite receiving favorable evaluations from students—were inconsistently executed and were hardly represented in top-tier platforms (Figure 2). This indicates that top-tier platforms emphasize structured gamification features that foster trust and organization, whereas mid- or lower-rank platforms depend on simple features that fail to maintain engagement.

These discrepancies directly influence educational outcomes. Students utilizing top-ranked platforms indicated elevated satisfaction, enhanced motivation, and prolonged engagement, attributable to the incorporation of organized and pedagogically coherent gamification features. The irregular or nonexistent utilization of essential features in lower-tier platforms correlates with diminished engagement and increased dropout rates.

Answer to RQ3: Higher-ranked platforms integrate core gamification features systematically—focusing on interactivity, rewards, previews, and clear course structures—which drives stronger engagement. By contrast, lower-tier platforms adopt fewer or less effective features, leading to weaker learner participation and reduced overall impact.

7.4 Hypothesis Validation

The hypothesis of this study proposed that the lack or inconsistent use of gamification elements would diminish learner engagement, whereas systematic incorporation of contextually relevant features would enhance participation. Our analyses support this hypothesis, but with important nuances.

First, the platform comparison and descriptive survey results (Table 4, Figures 1–2) clearly showed that platforms consistently implementing gamification features such as interactive assignments, lesson completion rewards, free previews, and detailed course outlines achieved higher student engagement. Conversely, mid- and low-tier platforms that lacked these elements exhibited weaker engagement, underscoring the central role of gamification in sustaining motivation.

Second, to evaluate whether demographic factors moderate these effects (RQ2), we conducted chi-square tests across year of study, major, and age group. Table 5 revealed no statistically significant subgroup differences (all $p > 0.05$). The lowest observed p -value was for Year $\times$ Course Outline preference ($X^2 = 15.23$, $p = 0.293$), which suggested minor variation across years but not enough to establish statistical significance. Other tests, such as Major $\times$ Leaderboard preference ($X^2 = 12.73$, $p = 0.623$) and Age $\times$ Achievement Animations ($X^2 = 1.88$, $p = 0.597$), likewise indicated broadly uniform

patterns. Logistic regression further confirmed that demographics did not reliably predict leaderboard preference (Pseudo- $R^2 = 0.0077$).

Collectively, these findings offer the central hypothesis: gamification features are critical for engagement, and their absence correlates with diminished motivation and weaker platform performance. However, the data also demonstrate that demographic segmentation (year, major, age) does not significantly alter feature preferences. This suggests that in the Bangladeshi context, cultural and contextual factors may matter more than individual demographic differences in shaping how students respond to gamification.

8 Conclusion

This study provides strong evidence that thoughtfully implemented gamification can enhance e-learning engagement in Bangladesh. We found that interactive, incentive-driven features combined with trust-building transparency are most effective. Platforms that offered small interactive tasks, completion rewards, free preview lessons, and detailed course outlines (all anticipated by over 80% of students) achieved the highest engagement. Conversely, systems lacking these elements saw diminished student motivation. Notably, features blending incentives with informational trust (e.g. rewards coupled with previews/outlines) showed the greatest impact on participation. Our analysis also uncovered mismatches between learner expectations and current platform designs, especially in lower-tier services. Overall, the results reaffirm our hypothesis and suggest that to maximize engagement, e-learning systems in Bangladesh should integrate student-valued gamification features in a culturally sensitive way.

9 Limitations Addressed

We surveyed 218 undergraduate and graduate students across more than ten Bangladeshi universities. This diverse sample (Table 1) ensures the findings reflect a broad range of experiences. The within-subjects survey design (all participants answered the same questions) enhanced consistency and reliability in measuring feature expectations. We categorized 24 e-learning platforms into four tiers (Top, High, Mid, Low) based on student-reported popularity. By focusing on locally relevant systems and eliminating foreign-only platforms from analysis, we ensured our conclusions address the Bangladeshi context. Each shortlisted feature was manually verified on every platform's interface, recording its presence or absence. We employed quantitative methods, including heatmap visualizations and significance testing, to link gamification features with engagement metrics. Heatmaps allowed us to quickly identify which features correlate with higher engagement across platform tiers, while statistical analysis (e.g. correlation) confirmed the strength of these relationships. By combining student survey data with platform UI analysis, this study offers a holistic, human-centered examination. Unlike prior work that may have focused solely on performance metrics, we emphasize user expectations and user experience in a cultural context.

10 Future work

Comparative studies could disentangle the effects of game elements from those of instructional quality or pedagogical style. The “dark

side” of gamification merits can be studied. Implementing our findings in practice—such as developing a Bangla-language gamified platform or LMS plugin—will test their real-world impact. For example, future work could prototype integrations with existing tools (like Kahoot) or extend local LMS with gamification modules. These applied studies would demonstrate the feasibility and benefits of our recommendations.

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